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SOP-09: MAINTENANCE

Zarewa Aluminium and Plastics Ltd — System of Operations v3.0

Department: Maintenance

System modules: Machines registry (via Operations/Settings), Maintenance plans and work orders

Primary roles: Maintenance Manager (desig_mm), Maintenance Technician (desig_mtech)

Supporting roles: finance_manager (capex, cost posting), operations_officer

1. PURPOSE AND SCOPE

Maintenance ensures production machinery remains operational, safe, and cost-effective. The ERP tracks machines, meter readings, preventive maintenance plans, corrective work orders, maintenance events, and cost lines linked to GL 5020 (Maintenance COGS).

2. MACHINE REGISTRY

2.1 Machine Records

Field

Machine ID

Name / description

Location

Fixed asset link

Status

2.2 Meter Logs

machine_meter_logs track usage: - Operating hours - Metres produced (where meter fitted) - Fuel consumption links - Used for preventive maintenance scheduling

2.3 Responsibilities

Role

Duties

Maintenance Manager (desig_mm)

Plans, work order approval, technician assignment

Maintenance Technician (desig_mtech)	Execute work orders, log events, parts usage
Production Manager	Report breakdowns; release machine for maintenance window

3. PREVENTIVE MAINTENANCE

3.1 Maintenance Plans (MXPL-YY-NNNN)

Recurring scheduled maintenance:

Plan Type

Daily

Weekly

Monthly

Quarterly

Procedure:

1. Create **Maintenance Plan** linked to machine.
 2. Set frequency, checklist items, estimated duration.
 3. System generates **Work Orders** when due.
 4. MM assigns technician; tracks completion.
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4. CORRECTIVE MAINTENANCE

4.1 Work Orders (MXWO-YY-NNNN)

Ad-hoc or plan-generated repair jobs:

1. **Report breakdown** — Production or Operations logs issue.
2. MM creates **Maintenance Work Order**:
 - Machine, fault description, priority
 - Link to material request if parts needed
3. Status: Opened → acknowledged → approved → closed
4. Technician executes; logs **Maintenance Event** (MXEV-YY-NNNN).

4.2 Maintenance Events

Completion record: - Date, technician, hours spent - Parts used (linked to stock or expense)
- Labour cost - Root cause notes

4.3 Cost Lines

maintenance_cost_lines classify under: - **Maintenance** expense → GL 5020 (COGS) - May link to **payment request** for external contractor invoices

5. MATERIAL REQUESTS FOR PARTS

When maintenance requires spare parts:

1. Create **Material Request** from work order
 2. Operations fulfils from store or Procurement raises PO
 3. Cost allocated to maintenance work order
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6. ASSET CUSTODY

Maintenance tools and portable equipment tracked via asset_custody_events:

Event

assign

transfer

confirm_present

report_missing

Linked to HR discipline and recovery schedules if staff negligence established.

7. GATE PASS EVENTS

Equipment leaving site: - Record gate_pass_events with authorisation - Security verification - Return confirmation on re-entry

8. FIXED ASSET LINKAGE

Production machines linked to fixed asset register: - Acquisition cost from capex PO or manual registration - Depreciation via monthly run (60-month useful life for plant) - Disposal through Accounting desk (SOP-03 \$8)

9. DOWNTIME REPORTING

On machine breakdown:

1. Production Manager stops job allocation to affected machine
 2. MM creates urgent work order (priority: Critical — 4hr SLA)
 3. Log downtime start/end on work order
 4. Production reschedule communicated to Sales via official notice or memo
 5. Post-incident review if downtime >4 hours
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10. MAINTENANCE KPIS

KPI

Planned maintenance completion

Breakdown response time

Mean time between failures

Maintenance cost as % of production COGS

Reports: maintenance cost lines in expenses pack; machine meter trends on Operations overview.

11. MONTHLY MAINTENANCE CHECKLIST

- ☐ All due preventive plans completed or rescheduled with reason
 - ☐ Open work orders cleared or escalated
 - ☐ Meter logs entered for all active machines
 - ☐ Asset custody verification for portable tools
 - ☐ Maintenance costs reviewed with Finance (GL 5020)
 - ☐ Critical spares stock level checked with Operations
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End of SOP-09. Cross-references: SOP-04 (Material requests), SOP-03 (Fixed assets), SOP-07 (Asset custody/discipline).